

GigaHands validation and QC flow, and why the experiments use different subsets
(an eligible pool is not the sample an experiment evaluated)

GigaHands released data
(RGB video, per-view 2D observations, provided 3D, COLMAP cameras)

camera / coordinate convention validation
(CAM-EXP-001)

diagnosis of PART of the error tail (CAM-EXP-001.1 / 001.2)
all-zero 2D pattern: 26,083 observations | left/right identity swaps within a view: 7,101
RGB segment matching | chosen-frame index audit against official source
NOT explained: 15,611 BAD_2D_GEOMETRY, 10,207 UNRESOLVED_INSUFFICIENT_GEOMETRY

independent multi-view reconstruction + leave-one-camera-out
(CAM-EXP-001.3; the provided 3D is never an input)
agreement with provided 3D: 3.9977 mm median | wrong-hand control: 192 mm

QC manifest - 108,240 hand observations labelled
PASS_STRICT 32,826 | PASS_SINGLE_HAND 16,005 | REVIEW 10,236 | EXCLUDE 49,173

A. CAMERA-CLEAN subset
(needs RGB + valid camera parameters)

175 of 200 views, 40 physical cameras
25 excluded: no usable RGB segment
hand annotation quality is NOT a filter here

-> Experiments 3 and 4

B. HAND-CLEAN / BIMANUAL-CLEAN ELIGIBLE POOL
(needs valid hand annotation;
bimanual needs BOTH hands)

16,413 frames over 154 views

-> the pool Experiment 2 draws from,
and later hand-aware work

stratified sample ACTUALLY EVALUATED in CAM-EXP-002

1,200 frames drawn -> 1,173 evaluated -> 2,210 hands
134 views, 37 physical cameras, 5 sequences
(dropped: 11 no-detection, 16 ambiguous association)